



VIT
CHENNAI



Neurox 3D Neurological Brain Segmentation

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MOTIVATION / INTRODUCTION

- Existing MRI diagnosis systems usually focus on only one neurological disease at a time.
- Doctors often need multiple separate tools for tumor, stroke, and Alzheimer analysis.
- Many AI models do not provide confidence or uncertainty information for clinical decisions.
- NeuroX was developed to create a single unified system for multi-disease detection, segmentation, and automated reporting.

SDG GOAL NUMBER: 3

SDG Goal 3 – United Nations Good Health and Well-Being

OBJECTIVES

- Develop a unified AI-based system for analysing brain MRI scans.
- Detect multiple neurological disorders including brain tumor, stroke, and Alzheimer's disease.
- Apply MRI preprocessing techniques such as normalization and skull stripping for standardized input.
- Extract deep volumetric features using 3D CNN and transformer-based architecture.
- Perform accurate lesion segmentation for abnormal brain region localization.
- Generate confidence-aware predictions and automated diagnostic reports.

SCOPE OF THE PROJECT

- NeuroX is designed to analyze brain MRI scans for multiple neurological disorder detection within a single AI framework.
- It supports tumor segmentation, stroke lesion identification, and Alzheimer's classification with confidence estimation.
- The system can be extended for clinical deployment, additional brain diseases, and hospital workflow integration.
- It also enables 3D visualization and automated report generation for easier clinical interpretation

METHODOLOGY

MRI Acquisition & Preprocessing

- Brain MRI scans are loaded, normalized, skull-stripped, and resized to 96×96×96 for uniform input.

Input Routing

- Different MRI modalities are routed based on disease type (tumor, stroke, Alzheimer's).

Feature Extraction

- A 3D CNN encoder extracts volumetric spatial features from MRI data.

Global Context Learning

- A Transformer bottleneck captures long-range anatomical relationships.

Disease Detection

- Separate prediction heads identify tumor, stroke, and Alzheimer's presence.

Lesion Segmentation

- U-Net decoder generates tumor and stroke lesion masks.

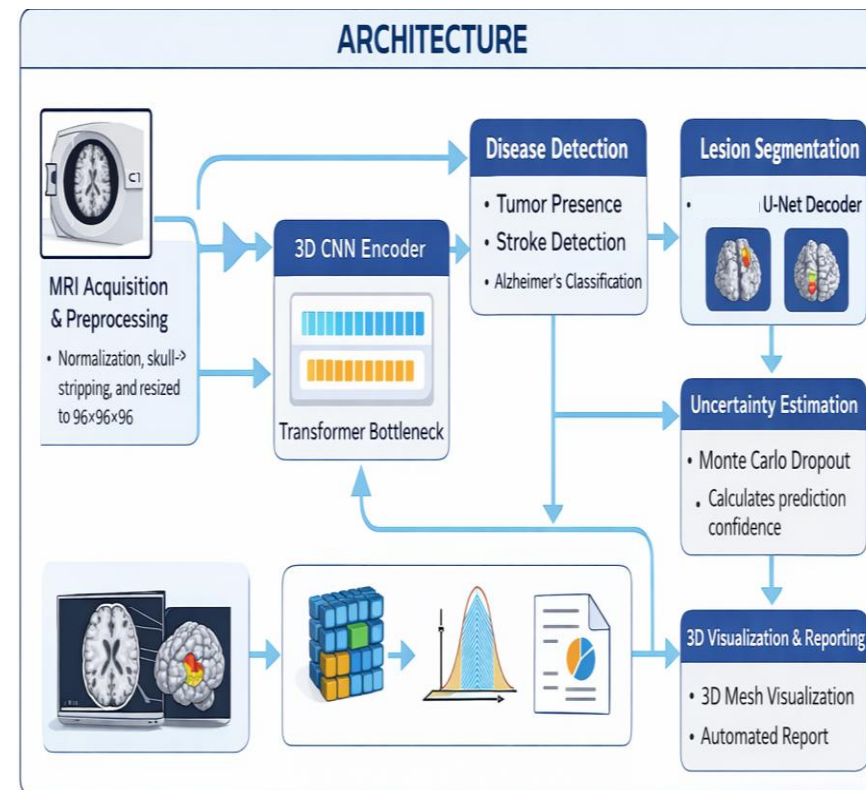
Uncertainty Estimation

- Monte Carlo Dropout calculates prediction confidence.

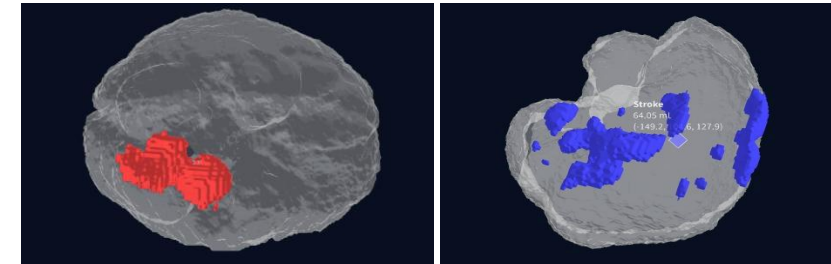
3D Visualization & Reporting

- Lesions are converted into 3D mesh models and an automated clinical report is generated.

ARCHITECTURE



RESULTS



CONCLUSION

- NeuroX successfully integrates tumor, stroke, and Alzheimer's detection into a single MRI analysis framework
- The proposed system combines 3D CNN, transformer bottleneck, and task-specific decoders for multi-disease prediction
- Accurate lesion segmentation was achieved for tumor and stroke cases
- Uncertainty estimation improved prediction reliability for clinical interpretation
- 3D visualization and automated reporting enhance practical usability
- The system demonstrates strong potential as an AI-assisted neuroimaging tool

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